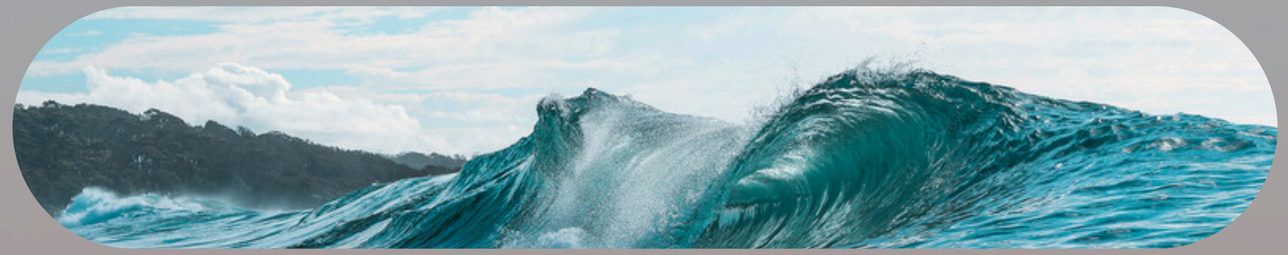
BRAINX PROGRAM

HYDRO TWIN

HYDRO





TWIN

Real-Time Water Intelligence for a Resilient Tunisia

Prepared by: balkis selmi & Nour elimen sbaiti

Before we start!



Tunisia-Climate change: Food security could be seriously impacted by 2050 and 2100News, Africa manager

What if engineers and farmers, could monitor and even predict floods, droughts, or water shortages before they happen?



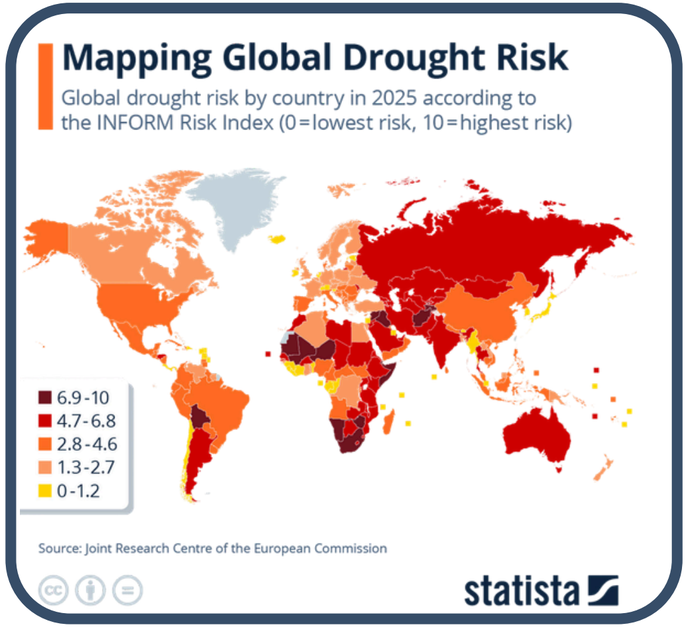
Damaged cars are parked in a flooded area in Hat Yai district, Songkhla province, Thailand [Athit Perawongmetha/Reuters]

In the past 12 months, has your local community experienced the following?

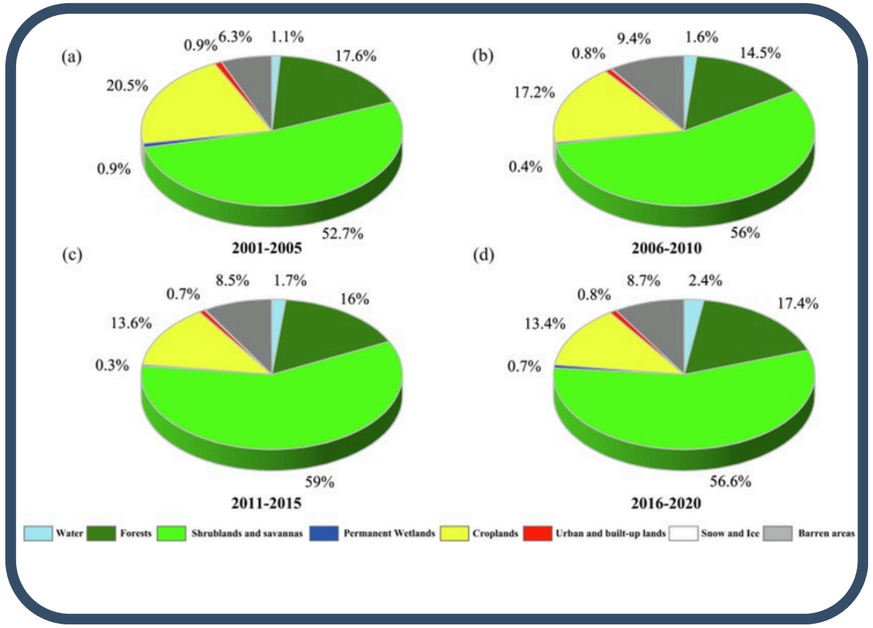
	Severe storms	Unusual heat	Drought	Flooding
Southwestern state	35%	53%	31%	15%
Rest of U.S.	50%	35%	15%	21%
Ten Across state	42%	56%	28%	17%
Rest of U.S.	47%	32%	15%	21%

Source: APM Research Lab analysis of McCourtney Institute's Mood of the Nation Poll, Jan. 11-17, 2024. N=1,000 U.S. adults age 18 or older; the overall margin of error is ±3.6 percentage points; larger for subgroups.

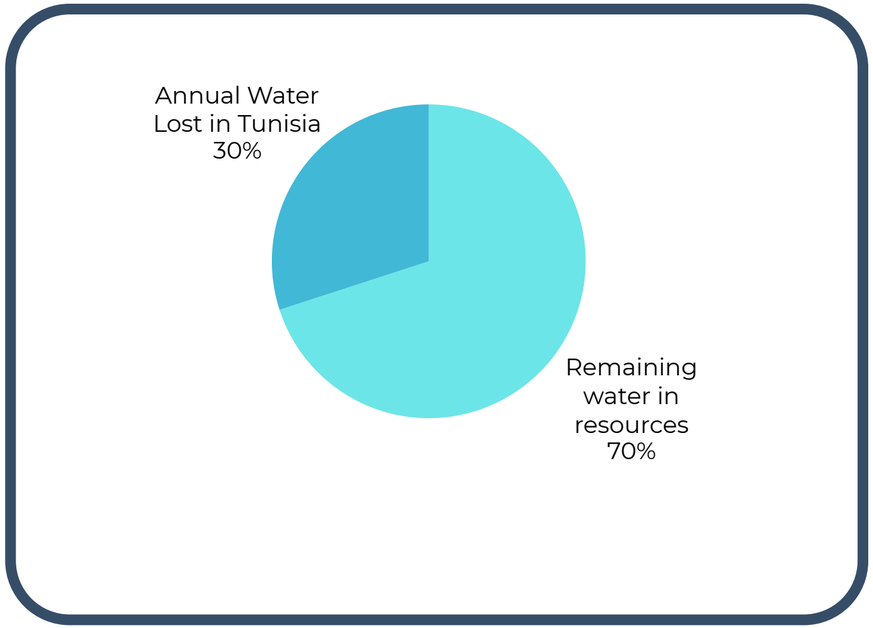
Unmonitored Water Conditions: A Growing Source of Major Issues



More than 1.6bn people, including almost 20% of the African population, will be exposed to severe and extreme droughts by 2050. / bne IntelliNews



Flooding are mainly affecting Croplands, forests, and vegetation globally, Journal of Hydrology



Water losses estimated at 30% of hydraulic resources, according to African Manager magazine

Our Product

AXIS 1

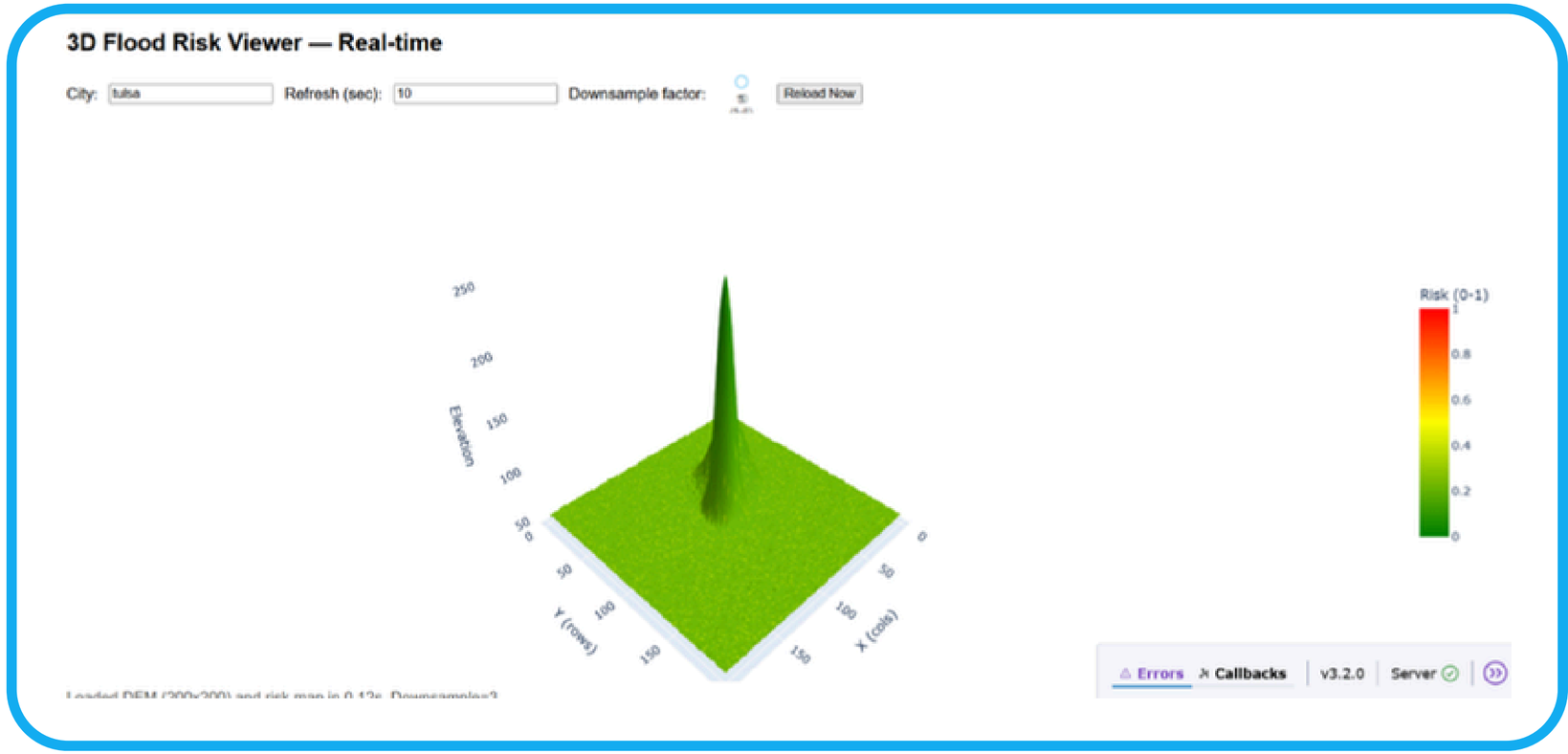
Risk Prediction and Visualization:
Drought, Flooding, Vegetation health, Urban damage...

AXIS 2

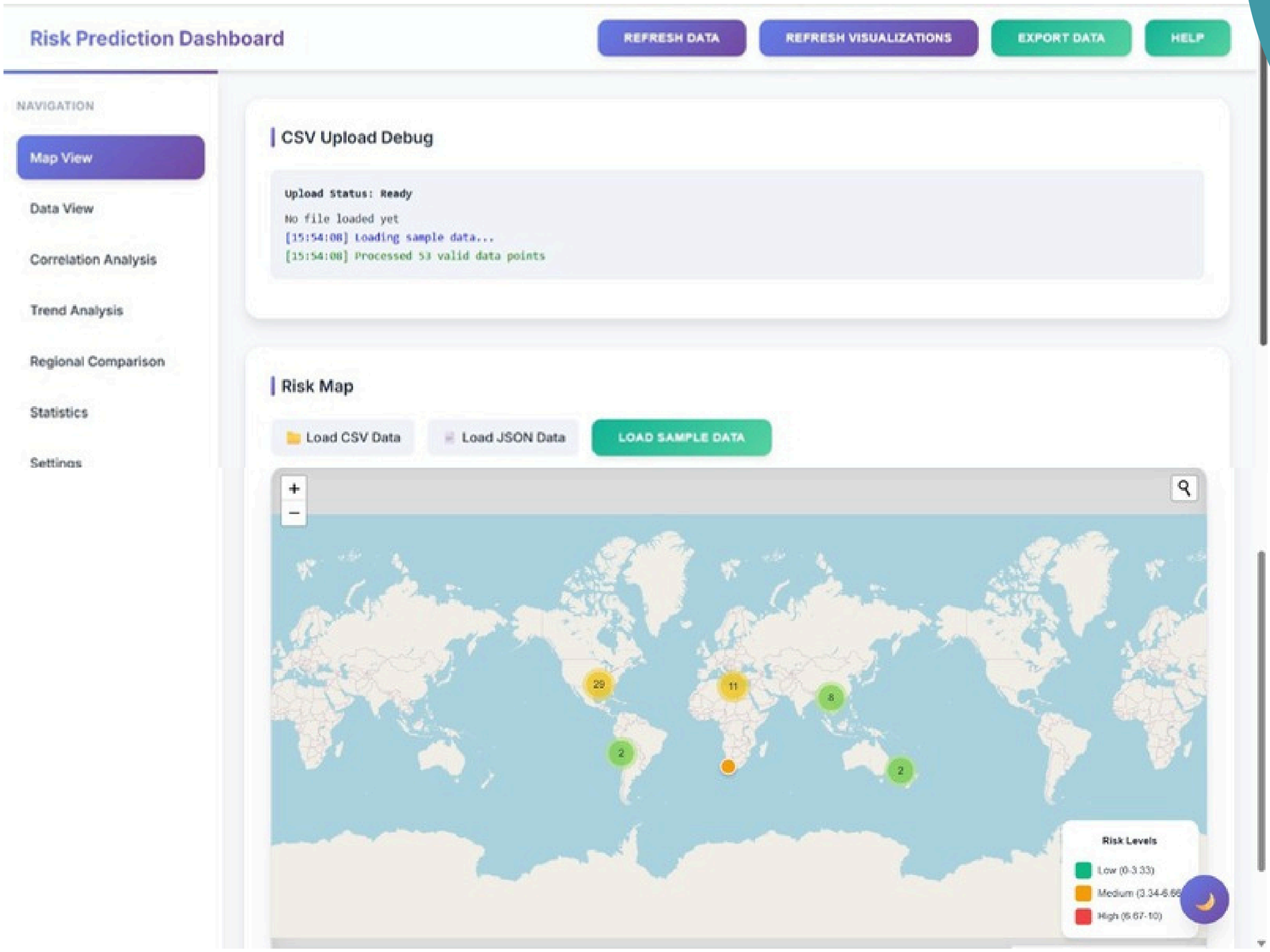
Providing solutions:
1- AI recommendation (simple surfaces)
2- onsite experts examination (complex surfaces

AXIS 3

Water cleaning systems depending on the property



First prototype of the 3D risk Visualization in Tulsa, Oklahoma

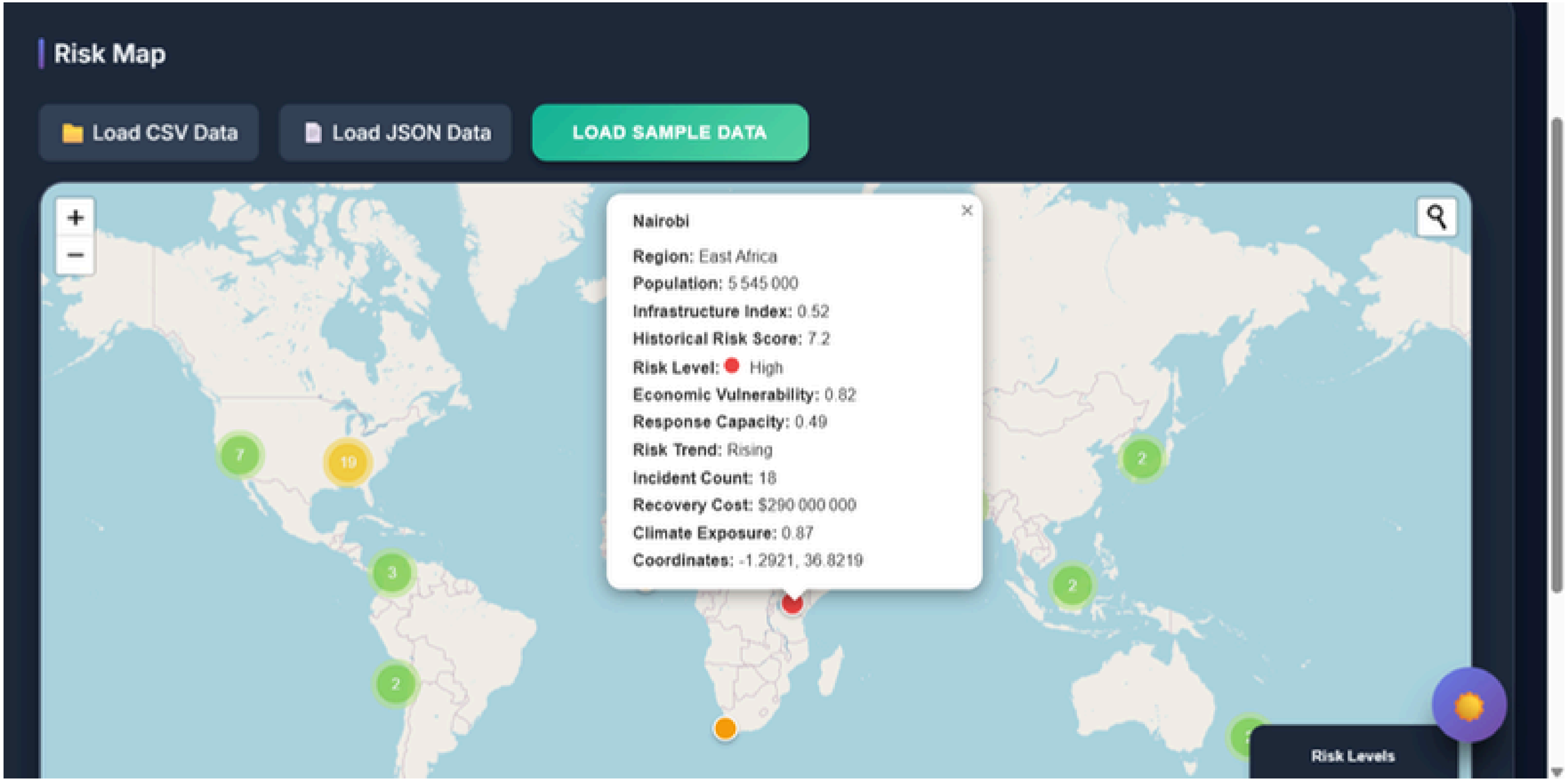

Risk Prediction Dashboard
REFRESH DATA
REFRESH VISUALIZATIONS
EXPORT DATA
HELP
NAVIGATION
Map View
Data View
Correlation Analysis
Trend Analysis
Regional Comparison
Statistics
Settings
CSV Upload Debug
Upload Status: Ready
No file loaded yet
[15:54:08] Loading sample data...
[15:54:08] Processed 53 valid data points
Risk Map
Load CSV Data
Load JSON Data
LOAD SAMPLE DATA
29
11
8
2
2
Risk Levels
Low (0-3.33)
Medium (3.34-6.66)
High (6.67-10)

Risk Prediction Dashboard

REFRESH DATA | REFRESH VISUALIZATIONS | EXPORT DATA | HELP

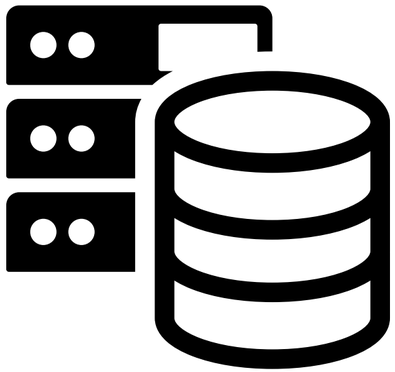
- Data View
- Correlation Analysis
- Trend Analysis
- Regional Comparison
- Statistics
- Settings

City	Region	Population	Infrastructure Index	Historical Risk Score	Economic Vulnerability	Response Capacity	Risk Trend	Incident Count	Recover Cost
New York, NY	North America	8 419 000	0.79	6.3	0.55	0.82	Stable	14	$520 00
Los Angeles, CA	North America	3 976 000	0.78	8.2	0.62	0.84	Stable	11	$430 00
Chicago, IL	North America	2 716 000	0.56	4.4	0.53	0.56	Rising	20	$389 07
Houston, TX	North America	2 328 000	0.72	4.6	0.69	0.85	Rising	13	$173 779
Phoenix, AZ	North America	1681000	0.9	4.8	0.34	0.59	Stable	19	$101295
Philadelphia, PA	North America	1584 200	0.72	4.0	0.72	0.58	Rising	20	$202 01
San Antonio, TX	North America	1547 200	0.76	6.3	0.32	0.64	Rising	2	$266

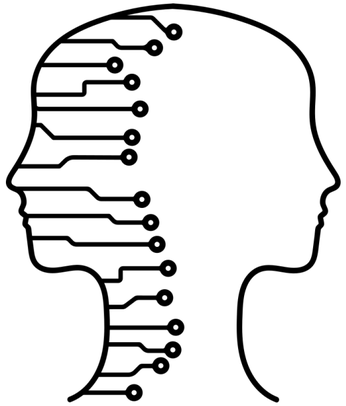

Risk Map
Load CSV Data
Load JSON Data
LOAD SAMPLE DATA
Nairobi
Region: East Africa
Population: 5 545 000
Infrastructure Index: 0.52
Historical Risk Score: 7.2
Risk Level: High
Economic Vulnerability: 0.82
Response Capacity: 0.49
Risk Trend: Rising
Incident Count: 18
Recovery Cost: $290 000 000
Climate Exposure: 0.87
Coordinates: -1.2921, 36.8219
7
10
3
2
2
2
Risk Levels

What's Special about our solution?

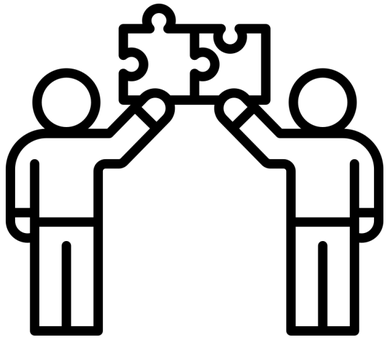
Unified Data Sources : Points for A Stronger Precision



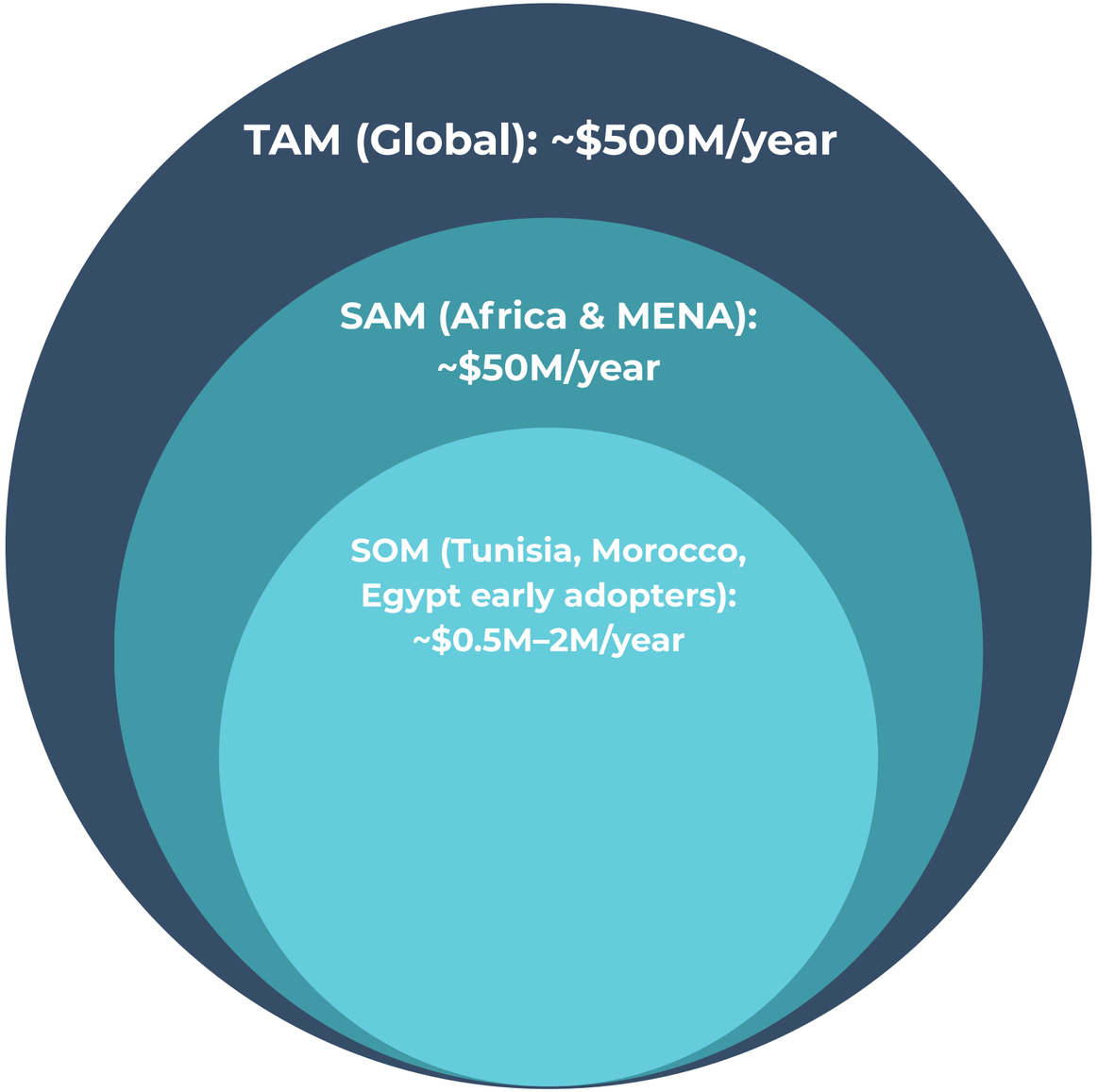
AI-Driven Risk Prediction



Fast Decision Making thanks to the AI recommendations and experts Team: Optimized Costs



Market Size / Opportunity

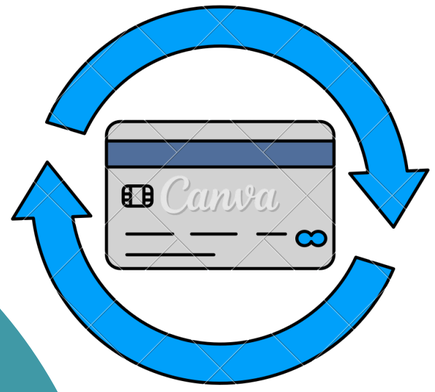


ADVANTAGES



Our Business Model

Annual platform and services Subscription for institutions & municipalities



Customized dashboards for agencies, labs, research centers



Future: Freemium mobile app for farmers & individuals



Our collaborations and potential partners

Research labs

GDA Sidi Amor

Laboratoire Eaux Usées et Environnement CERTE

Observatoire Tunisien de l'eau

Government

الجمهورية التونسية
وزارة الفلاحة والموارد المائية
والصيد البحري

وزارة التجهيز والإسكان والبنية التحتية
MINISTÈRE DE L'ÉQUIPEMENT DE L'HABITAT ET DE L'INFRASTRUCTURE
MINISTRY OF EQUIPMENT HOUSING AND INFRASTRUCTURE

Academic institutions

MSE

REPUBLIC OF TUNISIA
Ministry of Higher Education and Scientific Research

Competition

Current ecosystem

GDA Sidi Amor

Laboratoire Eaux Usées et Environnement CERTE

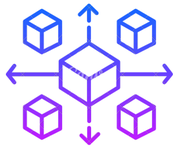
Observatoire Tunisien de l'eau

focused on research, advocacy, or limited monitoring.

Hydrotwin

- Real-time updates
- Predictive analytics
- Full integration of water risk indicators
- Scalable digital platform accessible to all users







BRAINX PROGRAM

الجمهورية التونسية
وزارة الفلاحة والموارد المائية
والصيد البحري

Team

Balkis Selmi

GEOMATICS ENGINEERING STUDENT

Nour Elimen Sbaiti

GEOMATICS ENGINEERING STUDENT

Zayneb Ben Rajab

GEOMATICS ENGINEERING STUDENT

Supervisors

Dr. Olfa Charfi

DOCTOR IN ELECTRICAL ENGINEERING AND MEMBER OF GREEN-TEAM LABORATORY (LR-INTEGRATED MANAGEMENT OF NATURAL RESOURCES : REMOTE SENSING, SPATIAL ANALYSIS AND MODELING)

Mr. Ali Ben Abbes

RESEARCHER IN COMPUTER SCIENCE & REMOTE SENSING APPLICATIONS

Our Road Map

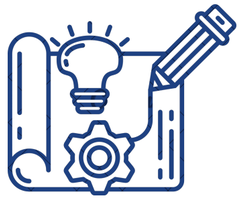
NOVEMBER,2025

Stakeholder interviews & requirements



NOVEMBER,2025

Platform prototype completed



STARTING FROM JANUARY,2026

Deploy pilot at Sijoumi Wetland



BY 2027

Scale across Tunisia → then MENA



From Challenge to Change

We are seeking:

- **Strategic partnerships**
- **Pilot sites with public water institutions**
- **Early investors to support deployment & scaling (data, sensors, aerial photogrammetry, and field truth for the correction**

OUR VISION:

“A world where every engineer, farmer, or policymaker has real-time water intelligence — shifting from crisis response to proactive water stewardship.”

Thank You!

